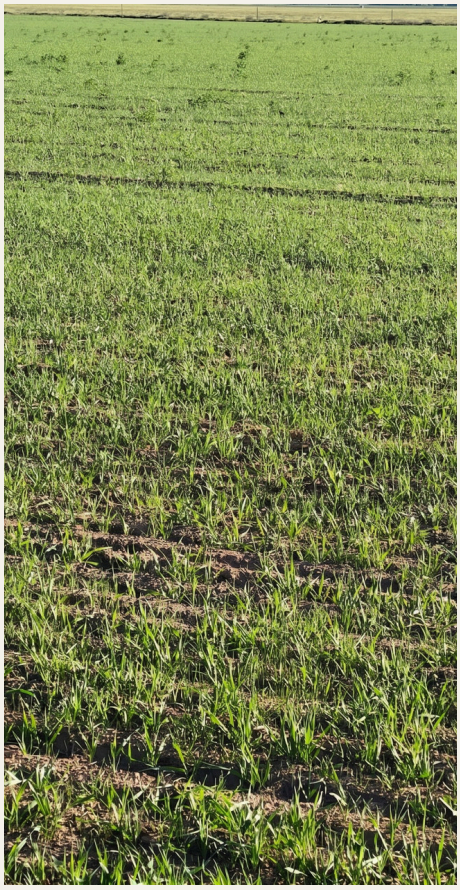


VEGETABLES



ORCHARDS



BROADACRE

MYCRO-TEC®

# DIGESTER

## Last season's residue, working for the next crop.

Four selected decomposer organisms, applied once onto crop residue after harvest. They begin the breakdown of stubble, prunings and crop trash so the carbon and nutrients held in it return to the soil ahead of the next crop.

**USED ON**
Crop residue in vegetables, orchards and broadacre

**APPLIED AS**
One foliar application onto the residue, after harvest

**CONTAINS**
Three fungi and one actinomycete, declared on the label

# Using Digester

## One application, after harvest, onto the residue and timed to moisture.



### VEGETABLES

Crop residue after harvest

| | |
|---|---|
| Foliar onto residue | **1–2 L/ha** |
| Heavy residue | **up to 3 L/ha** |
| Ammonium sulphate | **8 kg/ha** |
| Water | **100 L/ha** |
| Product : water | **1:50** |



### ORCHARDS

Prunings and leaf litter on the orchard floor. Not for nut trees.

| | |
|---|---|
| Foliar onto residue | **1–2 L/ha** |
| Heavy residue | **up to 3 L/ha** |
| Ammonium sulphate | **8 kg/ha** |
| Water | **100 L/ha** |
| Product : water | **1:50** |



### BROADACRE

Cereal, canola and legume stubble

| | |
|---|---|
| Foliar onto residue | **1–2 L/ha** |
| Heavy residue | **up to 3 L/ha** |
| Ammonium sulphate | **8 kg/ha** |
| Water | **100 L/ha** |
| Product : water | **1:50** |

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. One application after harvest; not for in-crop use.

## When it is worth doing

**Residue that will still be there at sowing**
Cereal, canola, legume, vegetable or orchard residue that would otherwise carry intact into the next crop.

**A moisture window you can hit**
Applied soon after harvest and close to forecast rain. The breakdown runs on residue moisture.

**Heavy or lignified loads**
Big stubble loads, or material that has sat through a dry summer and stalled.

**Where the basics are already in hand**
Residue breaks down fastest in a block where water, soil balance and nutrition are being managed. Digester is a step in that program, not a substitute for it.

**Hot, dry spells**
Activity falls away through prolonged heat and dryness. Line the application up with a viable moisture window.

**Time in the tank**
Mix and apply the same day. Overnight in nutrient the fungi grow mycelia and block nozzles.

**Nitrogen on its own**
Nitrogen feeds the biology already there; it does not add decomposers. The ammonium sulphate is a starter, not the treatment.

> If you run a nutrition program with us, Digester is the post-harvest step and the prescription blend at establishment follows it. If you don't, it stands on its own after harvest. Rate and timing depend on the residue load, the crop and the season. Your Hybrid-Ag agronomist can help decide whether Digester suits the block.

# What is in it and how it works

## Four organisms, each declared on the label, that between them work on every structural part of crop residue.

| | | |
|---|---|---|
| *Trichoderma harzianum*<br>FUNGUS | $>2\times10^{8}$<br>CFU/ML | Fast fungal coloniser of cellulose-rich residue; its enzymes begin opening the fibre. |
| *Aspergillus niger*<br>FUNGUS | $>2\times10^{7}$<br>CFU/ML | Produces organic acids and enzymes that work on the tougher, more lignified material. |
| *Streptomyces lydicus*<br>ACTINOMYCETE | $>2\times10^{7}$<br>CFU/ML | Adds bacterial decomposition alongside the fungi, so breakdown does not rely on one pathway. |
| *Penicillium radicum*<br>FUNGUS | $>1\times10^{7}$<br>CFU/ML | Broadens the decomposer team across the residue surface and the damp base. |

Colony-forming units per millilitre at production, as declared on the label. Three fungi and one actinomycete.

## What they act on

### Cellulose
The main plant fibre

### Hemicellulose
The supporting fibre network

### Lignin
The hard, resistant structure

### Pectin
The material that binds plant cells



### 1 Establishes at the base
The organisms take hold first where residue meets soil and moisture is held, then spread through damp stubble.



### 2 Opens the structure
Their enzymes and organic acids work on cellulose, hemicellulose, lignin and pectin, so the carbon locked in the fibre becomes available.



### 3 Turnover builds
The selected organisms and the soil's resident community keep working through the material ahead of the next crop.

**HOW YOU CHECK IT**

Digestion starts at the stubble base, where moisture sits. Treated residue pulls away from the soil more easily and the stalks soften and weaken as it runs. That is the one field check a grower can make without a lab.

*The roles described here are the known behaviour of these organisms on crop residue. They are not a measured result for Digester in a particular paddock or season.*

# Rates, mixing and handling

WHAT IS IN IT

CFU/mL at production · as declared on the label

| >2×10⁸ | >2×10⁷ | >2×10⁷ | >1×10⁷ |
|---|---|---|---|
| *Trichoderma harzianum* | *Aspergillus niger* | *Streptomyces lydicus* | *Penicillium radicum* |

## Application

| CROP GROUP | DIGESTER | STARTER | WATER |
|---|---|---|---|
| **Vegetables**<br>Crop residue after harvest | **1–2 L/ha**<br>up to 3 L/ha heavy residue | **8 kg/ha**<br>every application | **100 L/ha**<br>1:50 product : water |
| **Orchards**<br>Prunings and leaf litter on the orchard floor. Not for nut trees. | **1–2 L/ha**<br>up to 3 L/ha heavy residue | **8 kg/ha**<br>every application | **100 L/ha**<br>1:50 product : water |
| **Broadacre**<br>Cereal, canola and legume stubble | **1–2 L/ha**<br>up to 3 L/ha heavy residue | **8 kg/ha**<br>every application | **100 L/ha**<br>1:50 product : water |

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. One application after harvest; not for in-crop use.

## Mixing

Shake the drum well. Add Digester to cool, clean, non-chlorinated water first, then add the ammonium sulphate and stir thoroughly.

Strain as you pour. Keep nozzles and filters at 150 µm or larger.

Mix and apply the same day. Left in the tank overnight it grows mycelia and blocks nozzles.

Use only the approved mix of Digester, water and ammonium sulphate. No substitutes for the starter.

## Storage

Store at 5–40 °C, out of direct sunlight, drums upright. The lids are vented. Do not freeze.

Twelve months from the date of manufacture. Use opened drums promptly rather than keeping them.

Growth on top of the liquid is mycelia, a sign the biology is alive. Shake and use as normal.

## Safety

Keep out of reach of children. May irritate skin or eyes; wash after handling.

Wear gloves and eye protection when mixing. Follow the current label and safety data sheet.

Not certified organic. Allowable-input status is assessed case by case with your certifier.

## Talk to us about Digester

Tell us the crop, the residue load and when you expect rain. Your Hybrid-Ag agronomist can advise on timing and rate. Label and safety data sheet available on request.

**(03) 5722 7555** **hybridag.com.au**

Mycro-Tec® Digester is made in Australia by Great Southern Biology and supplied by Hybrid-Ag.

